1.Aim/Overview of the practical: 1.Write a python program to generate a simple bar graph using matplotlib. The graph should be properly labeled.

2.Write a python program to generate Pie-chart using matplotlib. The graph should be properly labeled.

3.Write a Python program to plot the function y = x2 using the matplotlib libraries.

2.Objective: To generate a simple bar graph with labeled axes and a title, pie chart with labeled sections and percentage values, plot the mathematical function y=x2y = x^2y=x2 using the matplotlib library.

3.Procedure/Algorithm/Code:

1. import matplotlib.pyplot as plt

```
# Data for the bar graph
categories = ['Category A', 'Category B', 'Category C', 'Category D']
values = [23, 45, 12, 30]
# Create the bar graph
plt.bar(categories, values, color='skyblue')
# Add labels and title
plt.xlabel('Categories')
plt.ylabel('Values')
plt.title('Simple Bar Graph')
plt.show()
```

2. import matplotlib.pyplot as plt

```
# Data for the pie chart
labels = ['Apple', 'Banana', 'Grapes', 'Orange']
sizes = [25, 35, 15, 25]
# Create the pie chart
plt.pie(sizes, labels=labels, autopct='%1.1f%%', startangle=90, colors=['red', 'yellow', 'green', 'orange'])
# Equal aspect ratio ensures that the pie chart is drawn as a circle.
plt.axis('equal')
# Add a title
plt.title('Fruit Distribution')
# Display the pie chart
plt.show()
```

3. import matplotlib.pyplot as plt

```
import numpy as np
# Create an array of x values
x = np.linspace(-10, 10, 100)
# Define y as x^2
y = x**2
# Create the plot
plt.plot(x, y, label='y = x^2', color='blue'
# Add labels and title
plt.xlabel('x')
plt.ylabel('y')
plt.title('Plot of y = x^2')
# Add a legend
```

```
plt.legend()

# Display the plot

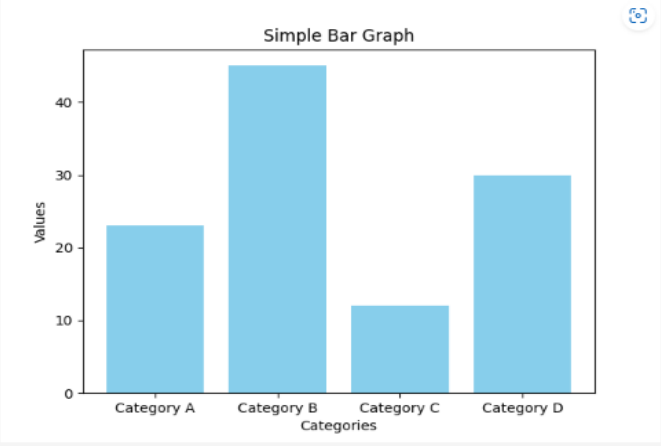
plt.show()
```

4.Output:

1.

Output

```
[Execution complete with exit code 0]
```



2.

Output

```
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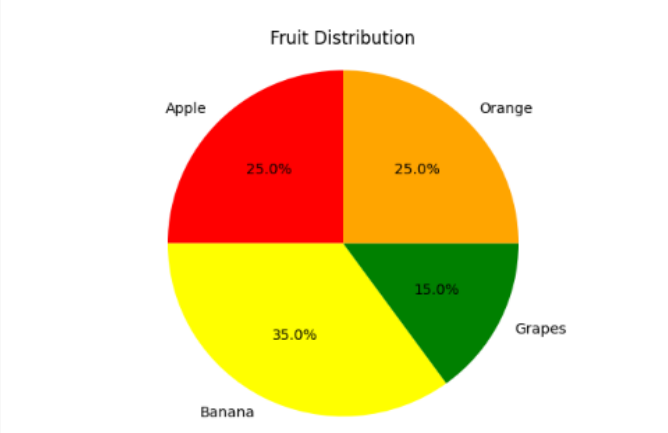
```



3.

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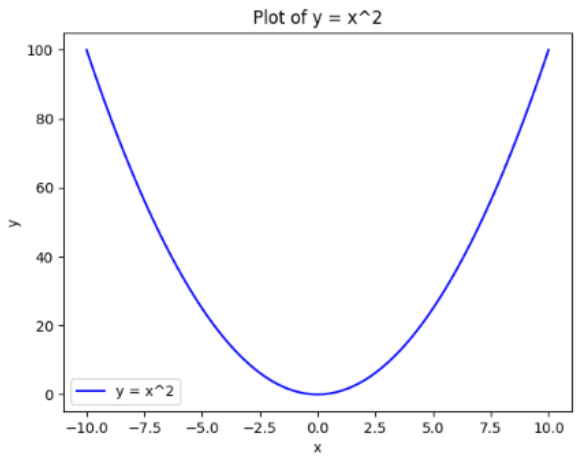
```



Learning outcomes (What I have learnt):

1. **Understanding the structure of a bar graph.**
2. **Pie Chart Using Matplotlib.**
3. **Plotting the Function y=x2y = x^2y=x2 Using Matplotlib.**